

SOSC 13200: Social Science Inquiry II

Section 6

University of Chicago

Winter Quarter 2025

Class details

Location: Gates-Blake 321

Meeting times: Tue/Thu 14:00 hrs – 15:20 hrs

Instructor

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Office hours: Mondays 12:00 hrs – 13:00 hrs

COURSE DESCRIPTION AND OBJECTIVES

In this course, you will learn to approach data thinking like a social scientist. That means thinking about where your data comes from, how it's measured, and assessing and describing relationships among variables. Throughout the course, you will be exposed to statistical tests and other quantitative methods used in social science research. You will also work on statistical software programming and data visualization.

By the end of this course, you will:

- Gain understanding of basic statistical concepts including sampling, descriptive statistics, and inferential statistics;
- Choose appropriate statistical tests for causality in terms of potential outcomes;
- Apply statistical concepts and methods to empirical investigations of core theoretical questions in the social sciences;
- Critically evaluate the use of these concepts and methods in the social science literature;
- Plan and conduct empirical studies of your own.

This session of the course focuses on the theme of the “credibility revolution” in the social sciences: a move toward empirical studies built with quality data, rigorous research design, and appropriate strategies for analysis. The impact of this work was recognized by the 2021 Nobel prize in economics, awarded to David Card, Joshua Angrist, and Guido Imbens. We will also consider the contributions of carefully designed survey and field experiments. You will also learn ways by which you will be able to enhance the rigor and replicability of your own research.

TEXTBOOKS

Probability & Statistics

Huntington-Klein, Nick. (2021). *The effect: An introduction to research design and causality*. Chapman and Hall/CRC. Online book: <https://theeffectbook.net/>

[For reference, not assigned] Wackerly, Dennis, William Mendenhall, and Richard L. Scheaffer (2014). *Mathematical statistics with applications*. Cengage Learning.

Coding in R

Verzani, John. *simpleR – Using R for Introductory Statistics*. Online book: <https://cran.r-project.org/doc/contrib/Verzani-SimpleR.pdf>

Wickham, Hadley, Mine Cetinkaya-Rundel, and Garrett Grolemund (2024). *R for Data Science (2e)*. O'Reilly Media. Online book: <https://r4ds.hadley.nz/>

MATHEMATICAL PROFICIENCY

Students need not have had prior exposure to probability and statistics, and this course will not use calculus or linear algebra to succeed in this class. Familiarity with these topics may be useful for some of the extra reference readings, but this is not required. At times, even if you have taken a stats class in high school or at the university, some of the ways that we will work with the concepts will be different from what you may have covered in that class.

STATISTICAL SOFTWARE

The statistical software used in this iteration of this class will be R, which we will use along with the RStudio interface. R is a statistical language and environment for data manipulation, calculation, and visualization; the software for R and Rstudio are free to download online. R can be a very flexible and powerful tool, and it is widely used in statistical and social science research, as well as in some industry settings. Instructors in other sessions of this course may use Stata, which is also widely used and relevant in particular in the fields of economics and public policy. In general, basic familiarity with both R and Stata is useful for research in the social sciences. However, in this version of the class, we will only use R. You are not assumed to have prior experience with R for this class.

COMPUTING

You will need access to a computer throughout the quarter with R and Rstudio installed. If you do not have access to a private computer to install this software, University library computers are equipped with Rstudio, and you should be able to save your assignments on Box while using these computers. If you have any issues with access to a computer, please reach out to me and we will find a solution.

ASSIGNMENTS

Homework consists of weekly assignments, submitted in R with compiled reports. During the last three weeks of the course, you will work on a final project, in which you will re-analyze a dataset of your choice, from published empirical academic work in the social sciences. In the final report, you will describe the data, hypotheses, what statistical tests are being implemented, and give your interpretation of results. Grade composition is:

- Homework: 60%
- Final project: 30%
- Participation: 10%

ACCOMMODATIONS

Please reach out to me directly if you would like to request accommodations for the course to better facilitate your learning. [Student Disability Services](#) is also available to provide you resources, support, and may provide approval for specific academic accommodations. Informing me in a timely manner will help me to ensure accommodations are met and I am able to implement an appropriate assessment of your learning.

ACADEMIC HONESTY AND AI ASSISTANTS

All students are expected to be familiar with and follow the University's [academic honesty policies](#). For this course, you may use generative artificial intelligence tools such as ChatGPT for generating code. No citation is required for code generated by AI, but you are responsible for understanding the code and ensuring it works correctly. You may also use AI tools for brainstorming, but you may not directly use any *text* generated from such tools in any assignments. Note that artificial intelligence may “hallucinate” information, so exercise extra caution and verify any information AI provides with reliable sources and proper citation practices (see discussions on AI hallucination in [this](#) and [this](#) New York Times articles).

COURSE OUTLINE

Week 1: Course introduction; appropriate questions

Class 1.1 Tuesday January 7

- Overview of course objective: how to formalize and test a theory with data and statistical tools
- Some motivating examples
- Introduction to coding: what the software will be, how you will download it, basics of setting yourself up with good file and directory hygiene.

Readings

- Tabarrok, Alex. [A Nobel Prize for the Credibility Revolution](#).

Learning R

- On your own, set up the swirl course on [R Programming with Email Notification](#).

Run the following code:

```
install.packages("swirl")
library(swirl)
install_course("R Programming E")
swirl()
```

Complete *Section 1: Basic Building Blocks* and *Section 2: Workspace and Files*

Class 1.2 Thursday January 9

What kinds of questions can we/might we want to ask in social science research? How do we start formalizing a question that is tractable and falsifiable? What population is the question in reference to?

Readings

- Holland, Paul W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*. Skip sections 5, 8.2.
- King, Keohane, and Verba (1994). Designing Social Inquiry: *Scientific Inference in Qualitative Research*. Princeton University Press. Chapter 1.

Homework 1

Due Monday 1/13 at 5pm

Install and set up statistical software and class working directory on your computer. Submit a compiled R document in pdf with your name and date, showing the file working directory.

Week 2: Summarizing data numerically and visually; probability intro

Class 2.1 Tuesday January 14

- Working with the Card and Krueger data set, summarizing data numerically and visually. Summaries of univariate data (sample mean, sample median, quantiles, sample variance, sample standard deviation).
- Graphical representation of different types of univariate data: histograms, density plots, boxplot, bar charts.
- Challenges with measurement.

Readings

- Adcock, Robert, and David Collier (2001). "Measurement validity: A shared standard for qualitative and quantitative research." *The American Political Science Review*.
- Card, David and Krueger, Alan B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*. *We'll come back to this paper again later this quarter to discuss the content and results; for this first pass, we'll just focus on the data.*

Learning R

- In swirl, complete *Sections 3: Sequences of Numbers, Section 4: Vectors, and Section 5: Missing Values*.
- *R for Data Science*, Chapter 1 and Chapter 2.

Statistical Reference [Optional]

- Huntington-Klein, Nick (2021). *The Effect*. Chapter 3.
- Wackerly et al. Chapter 1: What is Statistics?.

Class 2.2 Thursday January 16

- How do we think about what it means for something to be *random*? Why is this mathematical abstraction useful for social science research?
- Overview of discrete probability
- Conditional probability, Bayes Rule
- Why understanding conditional probability is essential to understanding social phenomena—and why relying on your intuition can be wrong

Readings

- Stark, P. B., Freedman, D. A., Mulargia, F., & Geller, R. J. (2003). What is the chance of an earthquake? *Earthquake science and seismic risk reduction*.
- Buchanan, M., (2007). [The Prosecutor's Fallacy](#), reproduced on Andrew Gelman's blog.
- Hill, R. (2004). [Multiple sudden infant deaths—coincidence or beyond coincidence?](#). *Paediatric and perinatal epidemiology*.

Learning R

In swirl, complete *Sections 6: Subsetting Vectors, Section 7: Matrices and Data Frames, Section 8: Logic, and Section 9: Functions*.

Statistical Reference [Optional]

- *Huntington-Klein, Nick (2021). The Effect. Chapter 4.*
- *Wackerly et al., Chapter 2: Probability; 2.1-2.4, 2.7, 2.10-2.13*

Homework 2

Due Monday 1/20 at 5pm

Submit a compiled R document in pdf with your name and date. Load in Card and Krueger dataset, and produce specified summary statistics and a plot.

Week 3: Probability as a model of the world

Class 3.1 Tuesday January 21

- Using Angrist & Krueger data to discuss identifying patterns in data, covariance, correlation.
- Education and earnings are correlated; preview correlation vs. causation with respect to the effect of education on earnings.

Readings

- Angrist, Joshua D., and Alan B. Krueger. "Does compulsory school attendance affect schooling and earnings?" *The Quarterly Journal of Economics* 106.4 (1991): 979-1014.

Statistical Reference [Optional]

- *Wackerly et al. Chapter 3: Discrete Random Variables and Their Probability Distributions; 3.1-3.3*
- *Wackerly et al. Chapter 5: Multivariate probability distributions; 5.1-5.4, 5.7 (ignore continuous distributions)*

Homework 3

Due Monday 1/27 at 5pm

Exercises in defining sample space, mapping events to probabilities. Using `sample()` to simulate random processes in R.

Class 3.2 Thursday January 23

- Continuous Variables

Statistical Reference [Optional]

- *Wackerly et al. Chapter 4: Continuous Variables and Their Probability Distributions*
- *Wackerly et al. Chapter 5: Multivariate probability distributions; 5.5, 5.6, 5.8 – 5.12*

Week 4: Inference

Class 4.1 Tuesday January 28

- Using Miguel and Kremer data, conditional means, difference in means.
- Data visualization: scatterplots, plotting conditional expectation function, faceting plots by category

Readings

- Ba, Bocar, Knox, Dean, Mummolo, Jonathan & Rivera, Roman. (2021). The role of officer race and gender in police-civilian interactions in Chicago. *Science*, 371(6530), 696-702.
- Miguel, Edward, & Kremer, Michael. (2004). Worms: identifying impacts on education and health in the presence of treatment externalities. *Econometrica*, 72(1), 159-217.
- [Optional] [Vox article](#) with additional content on the “Worm Wars” regarding replication issues with Miguel & Kremer

Learning R

- Verzani, 4: Bivariate Data, pp. 32-40

Statistical Reference [Optional]

- Huntington-Klein, Nick (2021). *The Effect*. Chapter 5.

Class 4.2 Thursday January 30

Using Butler and Broockman data, randomization inference, introduction to hypothesis testing, confidence intervals, p-values and statistical significance.

Readings

- Butler, Daniel M., & Broockman, David E. (2011). Do politicians racially discriminate against constituents? A field experiment on state legislators. *American Journal of Political Science*.
- Gerber, A. S., & Green, D. P. (2012). *Field experiments: Design, analysis, and interpretation*. WW Norton. Chapter 3: Sampling Distributions, Statistical Inference, and Hypothesis Testing, Sections 3.1-3.5, pp. 51-70.

Statistical Reference [Optional]

- S. Athey and G. Imbens. (2017). Chapter 3 - The Econometrics of Randomized experiments. In A. V. Banerjee and E. Duflo, editors, *Handbook of Field Experiments*, 1, pp. 73-140. North-Holland.

Homework 4

Due Monday 2/3 at 5pm

Data exploration, visualizing joint relationships.

Week 5: Hypothesis Testing

Class 5.1 Tuesday February 4

Sampling from a larger population; central limit theorem, one- and two-sample t-tests.

Readings

- Pager, Devah. The mark of a criminal record. *American Journal of Sociology* 108, no. 5 (2003): 937-975.

Statistical Reference [Optional]

- Wackerly et al. Chapter 7: Sampling Distributions and the Central Limit Theorem; 7.1-7.3
- Wackerly et al. Chapter 8: Estimation; 8.5-8.6

Class 5.2 Thursday February 6

Statistical power, effect sizes, sample size

Readings

- Coppock, Alexander. *10 Things to Know About Statistical Power*. EGAP, <https://egap.org/resource/10-things-to-know-about-statistical-power/>

Statistical Reference [Optional]

- Wackerly et al. *Chapter 10: 10.10 Power of Tests and the Neyman-Pearson Lemma*

Homework 5

Due Monday 2/10 at 5pm

Re-create the Pager data based on the description in the text. Formalize in words what hypotheses are being tested. Conduct the statistical analyses described. What is the reference population? Are these results generalizable? Why/why not?

Week 6: Bivariate regression: best linear predictor

Class 6.1 Tuesday February 11

- With Butler and Broockman data, regression as the best linear predictor, least squares visually demonstrated.
- Interpreting regression coefficients.

Readings

- Bueno de Mesquita, E., & Fowler, A. (2021). *Thinking clearly with data: A guide to quantitative reasoning and analysis*. Princeton University Press. Chapter 5: Regression for Describing and Forecasting.

Class 6.2 Thursday February 13

Inference for linear regression.

Linear regression with experiments and dummy variables; relationship to t-tests.

Readings

- NONE – Use this time researching data for your projects.

Statistical Reference [Optional]

- Huntington-Klein, Nick (2021). *The Effect*. Chapter 13.1; 13.2 through 13.2.5 inclusive only.
- Wackerly et al. *Chapter 11: Linear Models and Estimation by Least Squares; 11.1-11.4*

Homework 6

Due Monday 2/17 at 5pm

Replication of Butler and Broockman. What statistical tests are being used? How do you interpret the regression coefficients?

Get approval for your final project dataset.

Week 7: Multivariate regression: model building

Class 7.1 Tuesday February 18

Regression as linearization of conditional expectation function.

With Banerjee et al. data, interpreting regression coefficients (and standard errors) in multivariate regression.

Readings

- Banerjee, A., Duflo, E., Glennerster, R., & Kinnan, C. (2015). The miracle of microfinance? Evidence from a randomized evaluation. *American economic journal: Applied economics*, 7(1), 22-53.

Statistical Reference [Optional]

- Wackerly et al. Chapter 11: Linear Models and Estimation by Least Squares; 11.11-11.12

Class 7.2 Thursday February 20

When can we apply causal interpretations to regression coefficients from observational data?

Readings

- Card, D. and Krueger, A. B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*. *Now we'll come back to this paper and look at their approach to analysis.*
- Gerber, A. S., Green, D. P., & Kaplan, E. H. (2014). 1. The Illusion of Learning from Observational Research. In *Field experiments and their critics* (pp. 9-32). Yale University Press.

Homework 7

Due Wednesday 2/19 at 5pm

Final project part 1: data set selection and descriptive analysis.

Week 8: Regression: inference, challenges to inference, and external validity

Class 8.1 Tuesday February 25

Pitfalls of p-values and hypothesis testing; what's the alternative?

Readings

- Amrhein et al. (2019). Retire Statistical Significance. *Nature*.
- Gerber and Malhotra (2008). Do Statistical Reporting Standards Affect What is Published? Publication Bias in Two Leading Political Science Journals. *Quarterly Journal of Political Science*.
- Play around with "Hack Your Way To Scientific Glory," <https://projects.fivethirtyeight.com/p-hacking/>

Class 8.2 Thursday February 27

How our research design and data informs our analyses and generalizability

In-class time to discuss research projects

Readings

- Bohannon, John (2015). I Fooled Millions into Thinking Chocolate Helps Weight Loss. Here's How. *Gizmodo*.

Homework 8

Due Wednesday 2/26 at 5pm

Final project part 2: multiple linear regression and data visualizations.

Week 9: Declaring your hypotheses in advance; looking ahead

Class 9.1 Tuesday March 4

Filing pre-analysis plans

Wrap up and looking ahead

Readings

- Chen, Lula and Chris Grady. *10 Things to Know About Pre-Analysis Plans*, EGAP. <https://egap.org/resource/10-things-to-know-about-pre-analysis-plans/>
- Monogan, James (2015). "Research Preregistration in political science: The case, counterarguments, and a response to critiques." *PS* 425-429.

Class 9.2 Thursday March 6

TBD

Homework 9

Due Wednesday 3/5 at 5pm

Final project part 3: finalize report.